# Vision-Language Models for Interpretable Chest X-Ray Diagnosis Using the MIMIC-CXR Dataset

**Muhammad Taha Mukati**

**Student ID:** 730099705

## Abstract

Vision–language models (VLMs) have emerged as powerful tools for automated chest X-ray interpretation, offering improvements in diagnostic accuracy, multimodal reasoning, and clinical workflow support. However, their adoption in high-stakes medical settings remains limited by concerns around transparency and explainability. This literature review examines recent advances in medical VLMs, focusing on models such as CXR-CLIP, CheXagent, and BioViL-T, and evaluates how architectural design choices influence multimodal alignment, temporal reasoning, and interpretability. Particular attention is given to BioViL-T due to its clinically meaningful spatio-temporal attention mechanisms and its suitability for anatomically grounded explainability. Based on these insights, the project proposes a comprehensive plan to compare BioViL-T and CXR-CLIP using a curated subset of the MIMIC-CXR-JPG dataset, implementing a wide range of interpretability techniques which incorporates attention rollout, gradient-based attribution, concept-based methods, and counterfactual analyses. The planned evaluation aims to identify strengths, limitations, and potential failure modes in the interpretability of modern VLMs for chest X-ray diagnosis.

☑ **I have not used any GenAI tools in preparing this assessment.**

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**Signature:** Muhammad Taha Mukati

# 1 Introduction

Radiography remains one of the most ubiquitous and cost-effective imaging modalities in modern clinical assessment, providing non-invasive visualization of internal anatomical structures and enabling detection of diverse thoracic pathologies such as pneumonia, tuberculosis and cardiomegaly [1]. Among available imaging modalities, chest X- rays (CXRs), in particular, dominate as the most frequently performed imaging modality globally, according to a study published by the National Institutes of Health (NIH), over 70 million chest X-rays are performed annually in the United States alone, underscoring their critical role in global health care systems [2]. However, the interpretation of these images is inherently challenging due to subtle radiographic signs, inter-observer variability among radiologists, and the high volume of cases that strain healthcare resources [3]. This prevalent application highlights the integral role of CXRs in clinical workflows and highlights the demand for scalable, accurate, and interpretable automated diagnostic systems.

Over the past decade advances in deep learning—particularly in vision-language-models—have empowered medical practitioners to reduce cognitive workloads and shorten waiting lists by streamlining their workflows, thereby facilitating efficient and a reliable diagnosis of chest X-rays [[3], [4]]. Convolutional Neural Networks (CNNs) and, more recently, transformer-based architectures have become foundational in computer vision, underpinning automated systems for disease detection and localisation. CheXNet, CNN-based model, was capable of achieving radiologist-level performance in pneumonia detection from chest X-rays [5]. While these CNN- and transformer-based systems have demonstrated strong performance, their reliance on purely visual features limits their ability to capture the rich contextual information found in clinical narratives. This gap has catalyzed the emergence of vision-language models (VLMs), which jointly process images and text to deliver more holistic and interpretable diagnostic support [6].

Recent breakthroughs in vision-language models (VLMs) have addressed these limitations by enabling multi-modal learning that aligns visual and textual representations in a shared embedding space. VLMs, such as Open AI's Contrastive Language-Image Pre-training (CLIP), operate by training two neural encoders—one for images and one for text—on numerous of image-text pairs harvested from the internet to learn joint embeddings, enabling zero-shot learning capabilities that generalize across diverse tasks without extensive retraining [7]. In the biomedical domain, adaptations such as BioVIL and CXR-CLIP extend this framework by fine-tuning the vision and language encoder on domain-specific datasets, such as radiology reports and medical imaging data, when zero-shot performance is insufficient [8]. Due to the complex architectures of VLMs, these models often function as black-box systems, offering limited regarding the reasoning behind their decisions [9]. This lack of interpretability poses a significant challenge to clinical integration, as medical practitioners require not only accurate results but also interpretable outputs. In this literature review, we will examine these advances comprehensively by analysing the architectural foundations of leading VLMs, with particular focus on BioViL-T and CXR-CLIP. Additionally, explore the interpretability techniques developed for medical vision–language systems and provide a comparative analysis of these models to frame the methodological approach adopted in this project.

# 2 Literature Review

## 2.1 Background: Recent Advances in Vision-Language Models for Chest X-Ray Analysis

Traditional deep learning systems for chest X-ray interpretation–such as convolutional and transformer-based architectures like CheXNet–have demonstrated high diagnostic performance [5]. However, their reliance on large-labelled image datasets restricts their ability to incorporate contextual clinical information, often resulting in poor generalisation and challenges in differentiating between similar abnormalities. The emergence of VLMs has revolutionized chest X-they analysis by integrating visual X-ray data with textual radiology reports. This multimodal approach enhances diagnostic accuracy

and enables more interactive applications, such as visual question answering (VQA) and dynamic clinical querying, thereby offering a more holistic framework for chest X-ray analysis [10].

### 2.1.1 Foundation Models: CXR-CLIP

Over the past five years there has been rapid developments in Medical AI, particularly in Vision Language models. CLIP, a vision language model developed by Open AI, was trained on large-scale image-caption pairs from the internet to predict the correct matching batch of images and text [7]. However, due to its reliance on noisy, non-medical web data, CLIP is not directly suitable for medical imaging applications, where annotated datasets are limited and domain-specific terminology is essential. To address this limitation, CXR-CLIP extends the CLIP framework by introducing a prompt-based approach and making the model domain specific to Chest X-rays (CXR)s by leveraging datasets such as MIMIC-CXR [[11], [12]].

CXR-CLIP follows a dual-encoder architecture similar to CLIP [7], where visual and textual modalities are processed by separate encoders and projected into a shared latent embedding space for contrastive learning. The image encoder utilized ResNet-50 or a Swin Transformer as a backbone to extract global visual features, while the text encoder uses BioClinicalBERT [11]. Each encoder projects its features into a common 512-dimensional space, enabling contrastive learning between paired image-text samples. This model also offers a multi-view nature of CXR studies, by incorporating multiple images of frontal and lateral views in the chest x-ray section, as well as the findings and impression in the reports section.

CXR-CLIP achieved the best performance on the MIMIC-CXR dataset, with a top 1 retrieval recall of 21.6% and top 10 recall of 60.2%, outperforming other medical vision-language models. Its combination of CLIP loss, ICL and TCL allowed it to learn clinically relevant cross-modal representations that generalize well across radiology benchmarks.

### 2.1.2 State-of-the-Art: CheXagent

While CXR-CLIP achieves strong performance in zero-shot classification tasks, its contrastive dual-encoder architecture inherently restricts it to alignment and retrieval-based applications. In contrast, CheXagent, developed by researchers from Stanford University, represents a substantial advancement in medical vision–language modelling, built for real-world clinical deployment[13]. To train CheXagent, the team introduced a large-scale, instruction-based dataset called CheXinstruct. This dataset is one of the most comprehensive publicly available CXR datasets and comprises over 8.5 million training samples curated from 32 existing datasets, enabling CheXagent to acquire multi-task capabilities across image perception, image-text reasoning and clinical report generation[13].

Architecturally, CheXagent integrates a SigLiP-Large vision encoder with a Phi-2-based language decoder, jointly trained through a three-stage process involving medical text pre-training, vision-language alignment and instruction tuning [13]. This multi-stage training strategy significantly enhances CheXagent's generalisation and clinical adaptability compared to models such as Med-CLIP. In the initial stage, pre-training the language decoder on rich medical text by which the model develops a robust understanding of clinical semantics and diagnostic reasoning. The large-scale instruction tuning phase allows it to handle diverse downstream tasks such as disease classification, temporal progression analysis, report summarization, and visual question answering, all within a single unified model.

This approach not only improved its out-of distribution generalisation and cross-task transferability but also positioned CheXagent as one of the first vision-language models approaching clinical readiness. According to CheXbench, a benchmark suite for CheXagent, highlighted that the model achieved 0.993 accuracy, 0.90 in multi-disease classification, and 0.69 in temporal classification, consistently outperforming state-of-the-art models. Furthermore, the model was also evaluated in a clinical setting establishing that CheXagent contributed to improve CXR interpretation and writing efficiency in 23.8% of all evaluated cases, indicating its potential to serve as a copilot for radiologists [13].

## 2.2 The MIMIC-CXR Dataset

The MIMIC-CXR dataset (short for Medical Information Mart for Intensive Care – Chest X-rays) is a large, publicly available imaging dataset containing chest radiographs (X-rays) and associated radiology reports, designed for research in medical imaging and natural language processing, and multimodal learning [12]. The dataset comprises over 377,000 deidentified chest X-ray images from more than 65,000 patients, collected at the Beth Israel Deaconess Medical Centre (BIDMC) between 2011 and 2016 [[12], [14]]. Each imaging study in MIMIC-CXR typically includes frontal (anteroposterior or posteroanterior) and, where available lateral views of the chest [14]. Importantly, every image is linked to a free-text radiology report, which documents clinical findings, impressions, and occasionally the radiologist's level of diagnostic uncertainty.

## 2.3 BioViL / BioViL-T

### 2.3.1 Model Overview

Vision-language pre-training for bio-medical imaging, just like CXR-CLIP and CheXagent, requires a large dataset of paired radiology images and reports to learn joint multimodal representations to learn joint multimodal representations without requiring extensive manual annotation. This limits their ability to capture complex temporal and contextual information present in clinical narratives.

BioViL addresses this by introducing improved text semantic modelling through CXR-BERT, a specialized language model pre-trained on a vast dataset of radiology reports and fine-tuned to excel at interpreting domain-specific linguistic patterns, including negation (e.g. "no evidence of pneumonia), uncertainty (e.g., "possible mass"), and spatial references (e.g., "left lower lobe") that are ubiquitous yet challenging in clinical narratives [15]. This targeted optimization enables more nuanced and accurate textual embeddings, bridging the semantic gap between free-text reports and visual findings. BioViL-T extends this framework into the temporal domain by explicitly incorporating sequences of prior imaging studies alongside their corresponding reports. This integration allows the model to learn dynamic representation of disease progression such as the resolution of pulmonary edema or the development of fibrotic changes capturing critical temporal semantics that are absent in static, single time point alignments like CXR-CLIP and CheXagnet [16].

When compared to CXR-CLIP, which adapts CLIP's contrastive training to CXR data using radiology specific prompts, and CheXagent, which integrates multimodal reasoning with instruction-tuned large language models, BioViL focuses more on self-supervised alignment, interpretability, and efficient domain adaptation rather than broad generalisation and instruction following [[15], [16]].

### 2.3.2 Model Architecture

BioViL employs a dual-encoder design combining a ResNet-50 image encoder and CXR-BERT text encoder to learn joint representations of chest X-rays and their corresponding radiology reports. The image encoder, initialized through self-supervised pre-training on the MIMIC-CXR dataset using SimCLR framework with domain-specific augmentations, extracts spatially detailed [15]. These feature embeddings are passed through a two-layer projection head that maps them into a shared 120-dimensional latent space.

| Full word | ClinicalBERT | PubMedBERT | CXR-BERT |
|---|---|---|---|
| pneumonia | ✓ | ✓ | ✓ |
| opacity | op-acity | ✓ | ✓ |
| effusion | e-ff-usion | ✓ | ✓ |
| pneumothorax | p-ne-um-oth-orax | ✓ | ✓ |
| atelectasis | ate-lect-asis | ate-le-ct-asis | ✓ |
| cardiomegaly | card-io-me-gal-y | cardio-me-gal-y | ✓ |
| bibasilar | bi-bas-ila-r | bib-asi-la-r | ✓ |

Figure 1: Comparison of CXR-BERT with other text encoders

The textual counterpart, CXR-BERT, is a radiology-specialized language model pretrained on a large biomedical corpus to effectively model medical terminology (shown in figure 1), negation, and uncertainty expressions [15]. Its output embeddings are similarly projected into the same latent space through modality-specific projection head. BioViL's training objective combines a global contrastive alignments loss (InfoNCE) that maximizes agreement between paired-text samples while minimizing mismatched ones, and a masked language modelling loss that maintains semantic integrity of the text encoder during multi-modal training.

BioViL-T extends BioViL's functionality by utilising a multi-image encoder, designed to extract spatio-temporal features from sequential chest X-ray studies using a hybrid CNN-Transformer architecture [16]. This encoder combines the backbone ResNet-50 backbone with a transformer based temporal fusion module that model's dependencies across multiple time points. By incorporating temporal embeddings, the transformer learns to attend corresponding anatomical regions across the prior and current images, effectively capturing disease progression or resolution without explicit image registration. This design enables the model to distinguish static structural features from dynamic pathological changes through a feature decomposition mechanism [16]. This spatio-temporal representation is projected into a shared embedding space alongside the textual features from CXR-BERT, similarly like BioViL. This architecture allows BioViL-T to reason about changes in radiographs such as worsening consolidation or improving effusion making it highly suitable for interpretable temporal diagnosis on datasets like MIMIC-CXR.

## 2.4 Comparative Analysis of Vision-Language Models

| *Metric/Task* | *BioViL-T* | *CXR-CLIP* | *CheXagent* |
|---|---|---|---|
| ***Classification AUROC*** | 0.871 (zero-shot, pneumonia) | ~0.85 (zero-shot, multi-label) | 0.90 (multi-disease) |
| ***Report Gen (CheXbert F1 / ROUGE-L)*** | 0.317 / 0.296 | 0.25 / 0.20 | 0.249 / 0.403 |
| ***Temporal Acc/AUROC*** | 0.686 (macro-acc, progression) | N/A | 0.69 (temporal class.) |
| ***Interpretability*** | High (feature decomp., attention rollout) | Low (contrastive alignment) | Medium (report reasoning, fairness eval.) |

Figure 2: Comparison of BioViL-T, CXR-CLIP and CheXagent

Figure 2 shows that CheXagent achieves the strongest overall classification performance (multi-disease accuracy 0.903 on MIMIC-CXR), enabled by its decoder-based architecture and extensive instruction tuning. However, this comes with high computational cost and substantial model complexity. BioViL-T offers solid, semantically grounded performance (AUROC 0.871 on RSNA pneumonia, zero-shot), benefiting from its alignment between visual features and CXR-BERT text semantics. CXR-CLIP performs reasonably well in zero-shot settings (0.83 AUROC) but is limited by its reliance on handcrafted prompts and contrastive matching rather than explicit clinical reasoning.

BioViL-T provides the strongest and most clinically meaningful interpretability through spatial and temporal attention rollout, cross-modal attention between text tokens and image patches, and feature decomposition. These mechanisms produce anatomically grounded explanations aligned with radiological practice. CheXagent offers moderate interpretability via instruction-generated rationales, but explanations are narrative rather than spatially localised, limiting their transparency for diagnostic verification. CXR-CLIP provides the weakest interpretability restricted to similarity-based patch relevance maps due to the absence of cross-attention mechanisms.

Despite CheXagent's superior accuracy, it requires computational resources far beyond available capacity and its interpretability does not align with the project's focus on anatomically grounded, transformer-based explanations.

### 2.5 Interpretability Techniques

Interpretability in vision–language models (VLMs) for chest X-ray (CXR) diagnosis is essential for bridging the gap between high predictive performance and genuine clinical trust. Although many VLMs achieve impressive accuracy, their black-box nature can undermine adoption in high-stakes medical environments where explainability is critical for safe decision-making [9]. As a result, recent research has focused on developing interpretability techniques that offer post-hoc explanations, attention-based visualisations, and increasingly sophisticated counterfactual analyses aimed at revealing the model's underlying reasoning processes.

Gradient-based attribution methods, including established techniques such as Grad-CAM and Grad-CAM++, remain widely used for interpreting convolutional and transformer-based medical imaging models [17]. In CXR interpretation, Grad-CAM overlays these saliency maps onto the radiograph, helping to localise clinically significant findings such as consolidations, effusions, or cardiomegaly. This enables clinicians to verify that the model is focusing on anatomically plausible structures rather than artefacts or irrelevant regions [18]. Studies on large datasets like MIMIC-CXR demonstrate the value of such methods, with Grad-CAM improving alignment between predicted regions and expert annotations—achieving, for example, intersection-over-union scores around 0.45 for pneumonia localisation [19].

Transformer-based VLMs, by design, provide richer interpretability signals through attention rollout and cross-modal attention maps, which reveal how specific textual queries interact with visual features at the patch level [20]. Models such as BioViL-T incorporate temporal or hierarchical attention mechanisms that highlight evolving or multi-scale regions of interest, while CXR-BERT's self-attention structure elucidates how textual tokens such as "no pneumothorax" or "enlarged heart" influence downstream predictions [15]. These attention maps are typically computed during forward passes by aggregating or propagating attention weights across transformer layers, resulting in intuitive heat-maps that can be overlaid on radiographs.

Concept-based interpretability moves beyond pixel- or patch-level explanations by learning and visualising high-level clinical concepts that the VLM explicitly recognises, such as "increases cardiothoracic ratio" or "air bronchograms [21]. Methods like TCAV (Testing with Concept Activation Vectors) and its multimodal extensions train lightweight linear probes on user-defined concept examples to measure how sensitive a prediction is to each concept [22]. Recent works further combine concept probes with BioViL-T and CheXagent, enabling radiologists to query concept importance directly, yielding explanations that significantly outperform traditional heat-maps in clinician-rated plausibility on MIMIC-CXR [23].

Counterfactual reasoning methods simulate "what-if" scenarios by making minimal, targeted input changes that alter a model's prediction, revealing causal dependencies among visual and textual features[24]. In the context of CXR VLMs, approaches such as DiCE generate coherent and clinically plausible edits by leveraging generative models that preserve the overall realism of the radiograph while perturbing specific attributes [25]. When Applied to models like BioViL-T, these counterfactuals expose subtle failure modes—for example, cases where the model relies on rib shadows or imaging artefacts. Despite these advances, counterfactual interpretability in medical VLMs remains unexplored and is a promising for future research.

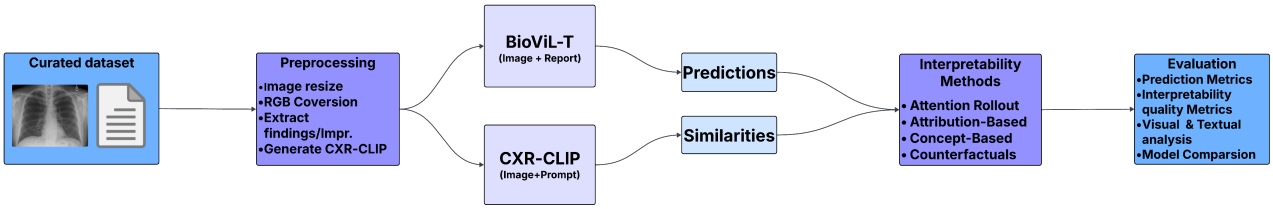
## 3 Project Specification

### 3.1 Aims and Objectives

The aim of this project is to investigate and evaluate a range of interpretable techniques for state-of-the-art vision-language models (VLMs) applied to chest X-ray diagnosis, with a primary focus on BioViL-T and a comparative baseline of CXR-CLIP. Using the MIMIC-CXR-JPG dataset, the project seeks to examine how architectural differences, multimodal alignment, and temporal modelling influence the clarity, reliability, and clinical relevance of the model. The objectives include:

- Curate a high-quality subset of the MIMIC-CXR-JPG dataset, ensuring images and reports are diagnostically meaningful, representative, and suitable for interpretability evaluation.
- Develop a unified inference pipeline (as shown in subsection 3.2) for BioViL-T and CXR-CLIP, incorporating image preprocessing, text handling, and prompt construction to enable consistent and reproducible prediction generation.
- Implement a comprehensive set of interpretability methods, including attention rollout, cross-modal attention maps, gradient-based attribution (e.g., Grad-CAM, Grad-CAM++), Integrated Gradients, concept-based approaches (TCAV), and counterfactual techniques.
- Evaluate model outputs and interpretability results using quantitative metrics and qualitative analysis to assess localisation quality, semantic alignment, and potential reliance on artefacts or non-clinical cues.
- Critically compare BioViL-T and CXR-CLIP across all interpretability outputs, analysing how architectural design influences explanation quality and identifying key challenges, limitations, and sources of error encountered throughout the project.

## 3.2 Model Pipeline




## 3.3 Dataset

This dataset was chosen due to its suitability for evaluating pre-trained VLMs such as BioViL-T and CXR-CLIP, which rely on paired image-text representations. As the aim of this project is to investigate interpretability. Therefore, a curated subset of MIMIC-CXR-JPG must be constructed to obtain a manageable, high-quality, and representative sample appropriate for generating meaningful interpretability outputs. This curated subset will prioritise images with clear anatomical visibility, complete and clinically relevant reports (for BioViL-T), and a balanced range of thoracic pathologies, while removing low-quality scans, duplicates, and studies lacking diagnostic content.

## 3.4 Model Inference

After curation, all images will undergo minimal preprocessing, including resizing normalisation to the expected model input dimensions, conversion to RGB format, and optional cropping to remove black borders. For BioViL-T, the corresponding report text will have to be extracted and formatted for CXR-BERT text encoder. For CXR-CLIP, manually constructed diagnostic prompts will be required to replace full reports, enabling zero-shot image-prompt comparison. Following preprocessing, the dataset will be passed through each model according to its input requirements. The BioViL-T encoder produces multimodal representations and predictions logits, whereas CXR-CLIP computes cosine similarity scores between image and prompt embeddings. The BioViL-T predictions and CXR-CLIP similarges of analysis.

## 3.5 Interpretability Method Implementations

### 3.5.1 Attention-based methods

Attention based interpretability methods will examine how models distribute focus across image regions. For BioViL-T, attention rollout will be used to aggregate transformer attention weights across layers, producing spatial and temporal attention maps that show which regions and prior studies influence predictions. Cross-modal attention visualisation will also be applied to reveal how specific CXR-BERT tokens (e.g., "effusion, "no pneumothorax") align with image patches. CXR-CLIP, lacking cross-attention due to its dual-encoder architecture, will be interpreted using similarity-based relevance mapping.

### 3.5.2 Attribution-based methods

Attribution methods will be used to analyse how input changes affect model outputs using gradient signals. Grad-CAM and Grad-CAM++ will be used to generate localised heatmaps for convolutional or transformer features, identifying regions most influential to a prediction. Integrated Gradients will provide a more robust, path-based attribution measure by integrating gradients between a baseline and the input image. These methods are architecture-agnostic and allow direct comparison between BioViL-T and CXR-CLIP.

### 3.5.3 Concept-Based Methods

Concept-based interpretability provides human-interpretable explanations linked to radiological concepts. TCAV (Testing with Concept Activation Vectors) is used to quantify how sensitive the model predictions are to learned concepts such as cardiomegaly and pleural effusion by analysing directional derivatives in the embedding space. Additionally, prototype retrieval identifies nearest-neighbour images or reports in the embedding space to expose concept-level similarity reasoning, particularly useful for CXR-CLIP's retrieval-oriented design.

### 3.5.4 Counterfactual Reasoning Methods

This project will implement several counterfactual reasoning techniques to assess model robustness, including latent-space counterfactual edits, attention-suppression counterfactuals, textual counterfactual query modifications for CXR-CLIP, and temporal counterfactual variations for BioViL-T. These approaches will be applied across both models to generate counterfactual outputs that support comparative interpretability analysis.

## 3.6 Evaluation Techniques

| **Evaluation Component** | **Implementation Notes** |
|---|---|
| **Localisation Quality (LQ)** | Heatmaps via Grad-CAM / Grad-CAM++ / attention rollout. Quantitative scoring via Pointing Game (no bounding boxes required). |
| **Semantic Alignment (SA)** | TCAV using model embeddings; cross-modal attention maps extracted directly from BioViL-T; similarity-based relevance for CXR-CLIP. |
| **Faithfulness (F)** | Integrated Gradients via Captum; deletion/insertion tests for causal verification; simple counterfactuals (masking, blurring, prompt edits). |
| **Clinical Plausibility (CP)** | Qualitative inspection; artefact-check heuristics (e.g., border or rib overfocusing); optional Likert-scale scoring. |

Table 1: Implementation difficulty of evaluation components for interpretability assessment of BioViL-T and CXR-CLIP.

# 4 Deliverables & Milestones

| Milestone | Deliverable |
|---|---|
| Dataset curation | Successfully create a subset of the MIMIC-CXR-JPG dataset. |
| Inference pipeline | Successfully implement the BioViL-T & CXR-CLIP pipeline. |
| Implement interpretability methods | Implement all the outlined interpretable techniques. |
| Evaluation results | Evaluate the interpretability methods for each model using suitable metrics. |
| Dissertation report | Critically report and analyse all implementations and findings. |

Table 2: Project milestones and deliverables

# 5 Project Timeline

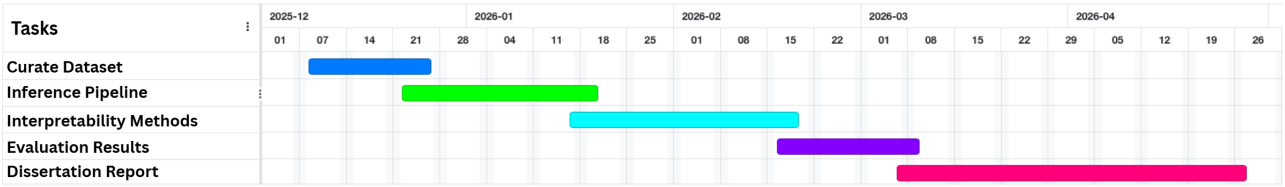



Figure 3: Illustrates the project timeline in accordance to the workload

- **Time Slippage:** ±7 days under mitigating circumstances.
- **Stretch Goal:** Develop and assess a custom hybrid interpretability method that combines cross-modal attention with counterfactual reasoning for BioViL-T

# 6 Ethics, Risks and Mitigations

The project uses the de-identified MIMIC-CXR dataset, which lowers privacy risks but still requires responsible handling. All data will be accessed under the MIMIC licence, stored securely, and used only for research, with no attempts at re-identification.

Ethical risks mainly relate to unreliable or biased explanations produced by VLMs. Models may focus on artefacts, reflect dataset imbalances, or generate attention maps that seem plausible but lack clinical fidelity. To mitigate this, the project applies multiple interpretability methods, evaluates faithfulness and localisation quality, and checks for artefact-driven behaviours. Comparing BioViL-T and CXR-CLIP further helps identify model-specific weaknesses.

There is also a risk that outputs may be misinterpreted as diagnostic. All findings will be explicitly framed as experimental and non-clinical. Reproducibility and technical integrity are supported through clear documentation, controlled preprocessing, and a consistent inference pipeline.

# 7 Conclusion

This review has highlighted how recent advances in vision–language models have improved chest X-ray interpretation through multimodal learning, temporal modelling, and increasingly sophisticated interpretability techniques. While models such as CXR-CLIP, BioViL-T, and CheXagent demonstrate strong performance, their transparency, reasoning processes, and clinical reliability remain key challenges. The project plan builds directly on these insights by developing a unified inference pipeline and implementing a comprehensive set of interpretability methods to evaluate explanation quality across models. By comparing BioViL-T and CXR-CLIP, the project aims to identify strengths, limitations in their explanations. The resulting analysis will support a clearer understanding of how architectural differences shape interpretability and will contribute to ongoing efforts toward clinically grounded medical AI systems.

## References


[1] C. Clinic, "A chest x-ray uses radiation to create an image of your heart, lungs and bones: A chest x-ray can diagnose health conditions like pneumonia or copd," https://my.clevelandclinic.org/health/diagnostics/10228-chest-x-ray, January 17 2017, accessed: 2025-11-17.

[2] L. Iyeke, R. Moss, R. Hall, J. Wang, L. Sandhu, B. Appold, E. Kalontar, D. Menoudakos, M. Ramnarine, S. P. LaVine, S. Ahn, and M. Richman, "Reducing unnecessary 'admission' chest x-rays: An initiative to minimize low-value care," *Cureus*, vol. 14, no. 10, p. e29817, 2022.

[3] V. Kulkarni, M. Gawali, and A. Kharat, "Key technology considerations in developing and deploying machine learning models in clinical radiology practice," *JMIR Medical Informatics*, vol. 9, no. 9, p. e28776, 2021.

[4] P. G. Anderson, H. Tarder-Stoll, M. Alpaslan *et al.*, "Deep learning improves physician accuracy in the comprehensive detection of abnormalities on chest x-rays," *Scientific Reports*, vol. 14, p. 25151, 2024.

[5] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning," 2017. [Online]. Available: https://arxiv.org/abs/1711.05225

[6] Q. Chen, R. Zhao, S. Wang, V. M. H. Phan, A. van den Hengel, J. Verjans, Z. Liao, M.-S. To, Y. Xia, J. Chen, Y. Xie, and Q. Wu, "A survey of medical vision-and-language applications and their techniques," 2024. [Online]. Available: https://arxiv.org/abs/2411.12195

[7] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," 2021. [Online]. Available: https://arxiv.org/abs/2103.00020

[8] M. Mistretta and A. D. Bagdanov, "Re-tune: Incremental fine tuning of biomedical vision-language models for multi-label chest x-ray classification," 2024. [Online]. Available: https://arxiv.org/abs/2410.17827

[9] Y. Liu, Y. Zhang, and S. Yeung-Levy, "Mechanistic interpretability meets vision language models: Insights and limitations," in *The Fourth Blogpost Track at ICLR 2025*, 2025. [Online]. Available: https://openreview.net/forum?id=pZqvfsUpeh

[10] I. Hartsock and G. Rasool, "Vision-language models for medical report generation and visual question answering: a review," *Frontiers in Artificial Intelligence*, vol. 7, Nov. 2024. [Online]. Available: http://dx.doi.org/10.3389/frai.2024.1430984

[11] K. You, J. Gu, J. Ham, B. Park, J. Kim, E. K. Hong, W. Baek, and B. Roh, *CXR-CLIP: Toward Large Scale Chest X-ray Language-Image Pre-training.* Springer Nature Switzerland, 2023, p. 101–111. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-43895-0_10

[12] A. L. Goldberger, L. A. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C. K. Peng, and H. E. Stanley, "Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. E215–E220, 2000.

[13] Z. Chen, M. Varma, J. Xu, M. Paschali, D. V. Veen, A. Johnston, A. Youssef, L. Blankemeier, C. Bluethgen, S. Altmayer, J. M. J. Valanarasu, M. S. E. Muneer, E. P. Reis, J. P. Cohen, C. Olsen, T. M. Abraham, E. B. Tsai, C. F. Beaulieu, J. Jitsev, S. Gatidis, J.-B. Delbrouck, A. S. Chaudhari, and C. P. Langlotz, "A vision-language foundation model to enhance efficiency of chest x-ray interpretation," 2024. [Online]. Available: https://arxiv.org/abs/2401.12208

[14] A. E. W. Johnson, T. J. Pollard, N. R. Greenbaum, M. P. Lungren, C. ying Deng, Y. Peng, Z. Lu, R. G. Mark, S. J. Berkowitz, and S. Horng, "Mimic-cxr-jpg, a large publicly available database of labeled chest radiographs," 2019. [Online]. Available: https://arxiv.org/abs/1901.07042

[15] B. Boecking, N. Usuyama, S. Bannur, D. C. Castro, A. Schwaighofer, S. Hyland, M. Wetscherek, T. Naumann, A. Nori, J. Alvarez-Valle, H. Poon, and O. Oktay, *Making the Most of Text Semantics to Improve Biomedical Vision–Language Processing*. Springer Nature Switzerland, 2022, p. 1–21. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-20059-5_1

[16] S. Bannur, S. Hyland, Q. Liu, F. Pérez-García, M. Ilse, D. C. Castro, B. Boecking, H. Sharma, K. Bouzid, A. Thieme, A. Schwaighofer, M. Wetscherek, M. P. Lungren, A. Nori, J. Alvarez-Valle, and O. Oktay, "Learning to exploit temporal structure for biomedical vision-language processing," 2023. [Online]. Available: https://arxiv.org/abs/2301.04558

[17] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, p. 336–359, Oct. 2019. [Online]. Available: http://dx.doi.org/10.1007/s11263-019-01228-7

[18] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Gupta, J. Patel, M. Gidwani, J. Adebayo, M. D. Li, and J. Kalpathy-Cramer, "Assessing the (un)trustworthiness of saliency maps for localizing abnormalities in medical imaging," 2021. [Online]. Available: https://arxiv.org/abs/2008.02766

[19] K. Xu, S. Ghosh, Z. Liu, C. Deible, and K. Batmanghelich, "Anatomy-guided weakly-supervised abnormality localization in chest x-rays," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, ser. Lecture Notes in Computer Science, L. Wang, Q. Dou, P. T. Fletcher, S. Speidel, and S. Li, Eds., vol. 13435. Cham: Springer, 2022, pp. 654–664.

[20] H. Chefer, S. Gur, and L. Wolf, "Transformer interpretability beyond attention visualization," 2021. [Online]. Available: https://arxiv.org/abs/2012.09838

[21] P. W. Koh, T. Nguyen, Y. S. Tang, S. Mussmann, E. Pierson, B. Kim, and P. Liang, "Concept bottleneck models," 2020. [Online]. Available: https://arxiv.org/abs/2007.04612

[22] B. Kim, M. Wattenberg, J. Gilmer, C. Cai, J. Wexler, F. Viegas, and R. Sayres, "Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav)," 2018. [Online]. Available: https://arxiv.org/abs/1711.11279

[23] A. M. Storås, M. Dreyer, F. Pahde *et al.*, "Exploring the clinical value of concept-based ai explanations in gastrointestinal disease detection," *Scientific Reports*, vol. 15, p. 28860, 2025.

[24] R. Guidotti, "Counterfactual explanations and how to find them: Literature review and benchmarking," *Data Mining and Knowledge Discovery*, vol. 38, pp. 2770–2824, 2024.

[25] R. K. Mothilal, A. Sharma, and C. Tan, "Explaining machine learning classifiers through diverse counterfactual explanations," in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, ser. FAT* '20. ACM, Jan. 2020, p. 607–617. [Online]. Available: http://dx.doi.org/10.1145/3351095.3372850